🧪 LAB 4 Simulated Internet (10.10.0.0/16) & Private LANs with Stateful vs Stateless Services

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🎯 LAB GOALS

1. **Network Setup:** Create two local labs (A & B) connected via a simulated "public/internet" network (10.10.0.0/16)
2. **NAT + Routing:** Configure static routes and NAT so local traffic reaches remote services
3. **Application Deployment:** Use `mockup-infra` from your repository to deploy stateless and stateful APIs
4. **Testing:** Compare how sessions behave locally vs across the simulated internet

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สมาชิก

673380056-7 รพีภัทร แก้วคูณ (ไผ่) sec.2

673380068-0 สุชาดา ศรีจักรโคตร (เค้ก) sec.2

673380286-0 พิไลพร คำเวียง (เอวา) sec.2

673380297-5 หทัยพัทธ วิสุทธิธรรม (แก้ม) sec.1
673380299-1 เจตนิพิฐ แก้วทรัพย์ (ลูกแก้ว) sec.1

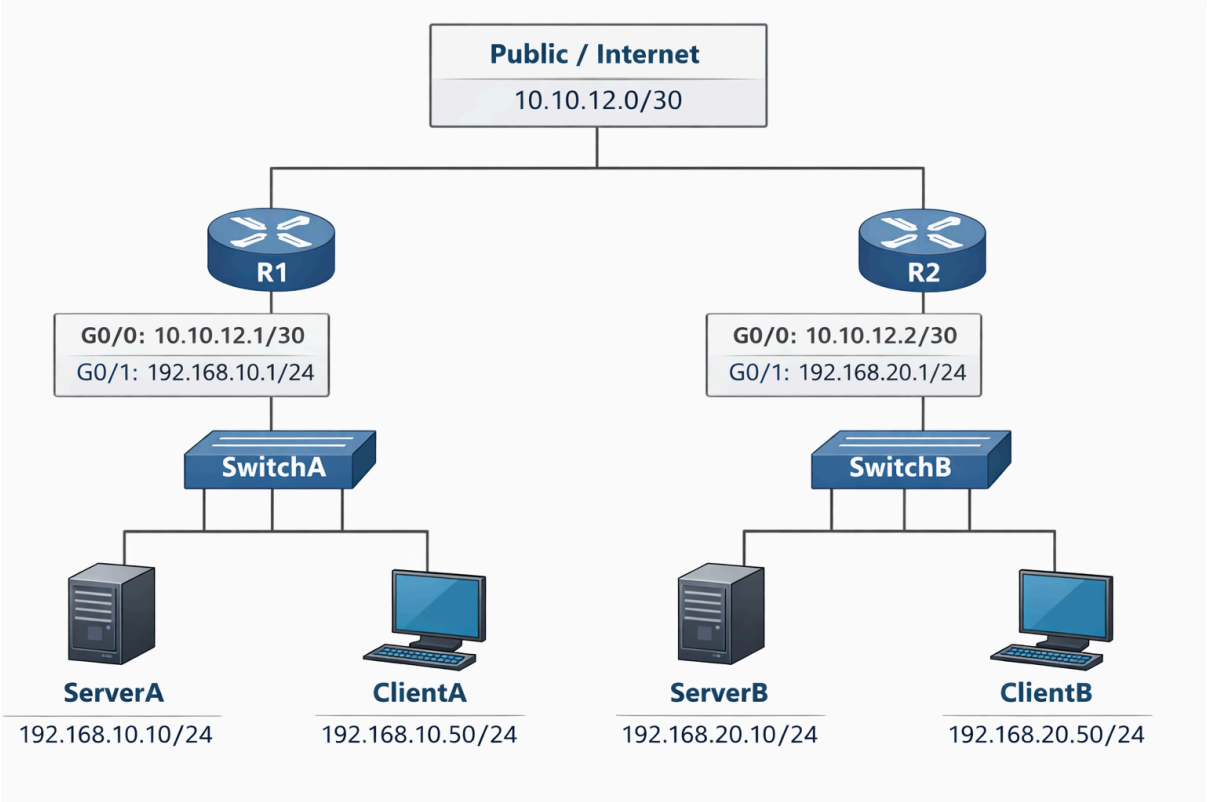
👥 TEAM BREAKDOWN (5–6 students)

Member	Role	Count	Focus
รพีภัทร พิไลพร	Network Architects	2	Router configs, Cisco CLI
สุชาดา หทัยพัทธ	IaC/DevOps	2	VS Code, mockup-infra
เจตนิพิฐ	QA/Test	1	Server/Client scripting, session behavior

🗺️ UPDATED NETWORK TOPOLOGY

```python
                10.10.12.0/30 "Public/Internet"
             ┌──────────────────────────┐
             |                          |
         R1-G0/0 (10.10.12.1)      R2-G0/0 (10.10.12.2)
             |                          |
   Private LAN A (192.168.10.0/24)     Private LAN B (192.168.20.0/24)
             |                          |
         SwitchA                      SwitchB
             |                          |
 ServerA (mockup)            ServerB (mockup)
 ClientA                         ClientB
```

We treat 10.10.12.0/30 as a **simulated internet** for the purposes of NAT + public reachability.

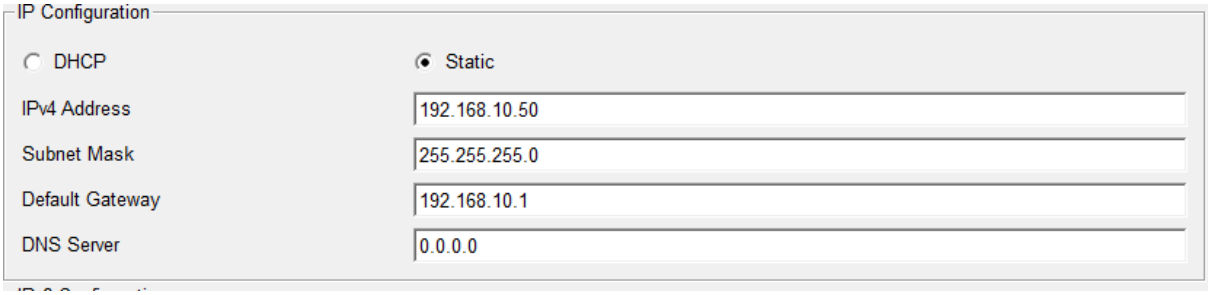


📍 ADDRESS TABLE

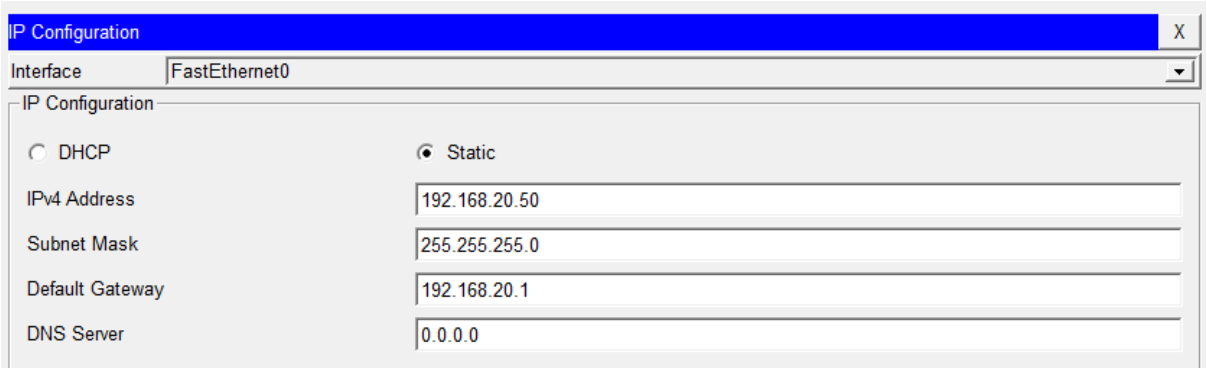
Component	Interface	IP	Mask	Role
R1	G0/0	10.10.12.1	/30	Public
R1	G0/1	192.168.10.1	/24	LAN A Gateway
ServerA	eth0	192.168.10.10	/24	Mockup infra A
ClientA	eth0	192.168.10.50	/24	Tester
R2	G0/0	10.10.12.2	/30	Public
R2	G0/1	192.168.20.1	/24	LAN B Gateway
ServerB	eth0	192.168.20.10	/24	Mockup infra B
ClientB	eth0	192.168.20.50	/24	Tester

จำลองเครือข่ายบน Cisco Packet Tracer

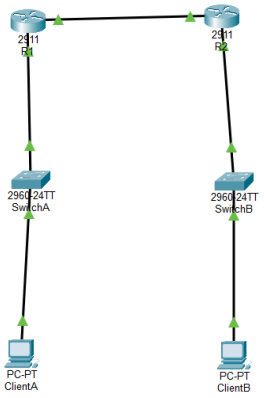
Client A :



Client B :



ทำการ Config IP Client A และ Client B



🔧 PART 1 — CISCO-STYLE NETWORK CONFIGURATION

We will configure static routing + NAT to allow cross-site reachability.

🧱 R1 CONFIGURATION (LAN A)

```
None
enable
conf t

hostname R1

interface g0/0
 ip address 10.10.12.1 255.255.255.252
 ip nat outside
 no shutdown

interface g0/1
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 no shutdown

ip route 192.168.20.0 255.255.255.0 10.10.12.2

access-list 10 permit 192.168.10.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload

end
write memory
```

ทำการ Config Router 1 :

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#inter
R1(config)#interface g0
R1(config)#interface g0/0
R1(config-if)#ip address 10.10.12.1 255.255.255.252
R1(config-if)#ip nat outside
R1(config-if)#no shut down

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
```

```
R1(config)#inter
R1(config)#interface  g
R1(config)#interface  g0/1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#ip nat inside
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

```
R1(config)#ip route 192.168.20.0 255.255.255.0 10.10.12.2
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#ip nat inside source list 10 interface g0/0 overload
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
```

R1 Config : ✅

🧱 R2 CONFIGURATION (LAN B)

```
None
enable
conf t

hostname R2

interface g0/0
 ip address 10.10.12.2 255.255.255.252
 ip nat outside
```

```
 no shutdown

interface g0/1
 ip address 192.168.20.1 255.255.255.0
 ip nat inside
 no shutdown

ip route 192.168.10.0 255.255.255.0 10.10.12.1

access-list 10 permit 192.168.20.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload

end
write memory
```

ทำการ Config Router 2 :

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface g0/0
R2(config-if)#ip address 10.10.12.2 255.255.255.252
R2(config-if)#ip nat outside
R2(config-if)#no shutdown


R2(config)#interface g0/1
R2(config-if)#ip address 192.168.20.1 255.255.255.0
R2(config-if)#ip nat inside
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

R2(config)#ip route 192.168.10.0 255.255.255.0 10.10.12.1
R2(config)#access-list 10 permit 192.168.20.0 0.0.0.255
R2(config)#ip nat inside source list 10 interface g0/0 overload
R2(config)#
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
```

R2 Config : ✅

🧪 ROUTING & NAT VERIFICATION

On R1 & R2:

```
None
show ip route
show ip nat translations
```

On R2 :

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.12.0/30 is directly connected, GigabitEthernet0/0
L       10.10.12.2/32 is directly connected, GigabitEthernet0/0
S    192.168.10.0/24 [1/0] via 10.10.12.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1


R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.12.0/30 is directly connected, GigabitEthernet0/0
L       10.10.12.2/32 is directly connected, GigabitEthernet0/0
S    192.168.10.0/24 [1/0] via 10.10.12.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1

R2#show ip nat translations
```

From ClientA:

```
None
ping 192.168.20.10  ← Remote server reachable
```

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Ping จาก ClientA ไปยัง ClientB ไม่สำเร็จ

💻 PART 2 — IaC + MOCKUP-INFRA DEPLOYMENT (mockup-infra/README.md)

We will now set up mockup infra using the paths you specified:

```
None
automation/mockup-infra
automation/week02-stateless-stateful/phase1-mockup
```

🛠️ CLONE REPO

```
Shell
git clone https://github.com/boonsup/automation.git
cd automation
```

🧰 VS CODE — OPEN PROJECT

In VS Code, open the `automation` folder.

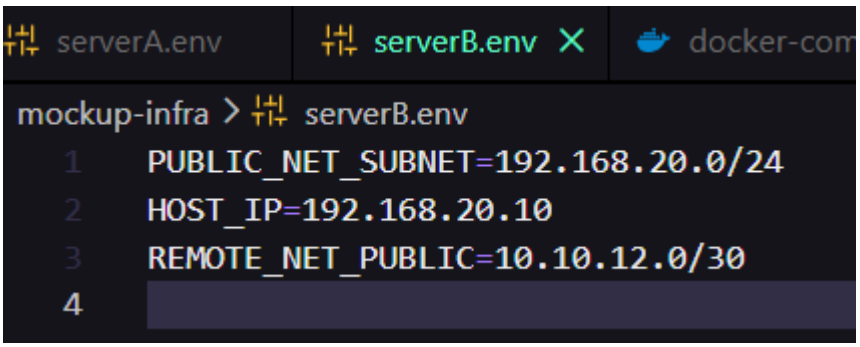
---

clone project และเข้าสู่โฟลเดอร์ `automation` สำเร็จแล้ว ดำเนินการต่อไปดังนี้

📄 CREATE SERVER-SPECIFIC ENV FILES

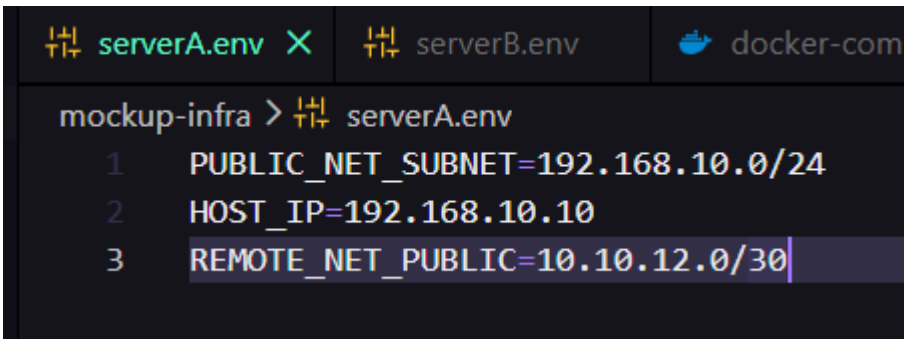
ServerA (`ServerA.env`)

```
None
# Private LAN A
PUBLIC_NET_SUBNET=192.168.10.0/24
HOST_IP=192.168.10.10
REMOTE_NET_PUBLIC=10.10.12.0/30
```



ServerB (`ServerB.env`)

```
None
# Private LAN B
PUBLIC_NET_SUBNET=192.168.20.0/24
HOST_IP=192.168.20.10
REMOTE_NET_PUBLIC=10.10.12.0/30
```



ทำการเซ็ต env เรียบร้อย✅

🚀 DEPLOY MOCKUP-INFRA

ใช้ PowerShell (Window)

PowerShell (Windows Admin)

```None
cd mockup-infra
python manage.py init
python manage.py deploy --profile week02-stateless-stateful
```

Bash (Linux)

```Shell
cd mockup-infra
chmod +x scripts/*.sh
./scripts/init-env.sh
python manage.py deploy --profile week02-stateless-stateful
```

```
PS C:\Users\User\Downloads\automation-main\automation-main\mockup-infra> wsl -d Ubuntu
🔐 [Layer 5/6 - Session] Generating TLS Certificates...
/mnt/c/Users/User/Downloads/automation-main/automation-main/mockup-infra/manage.py:118: DeprecationWarning: datetime.datetime.utcnow() is deprecated and schedu
led for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  .not_valid_before(datetime.utcnow())
/mnt/c/Users/User/Downloads/automation-main/automation-main/mockup-infra/manage.py:119: DeprecationWarning: datetime.datetime.utcnow() is deprecated and schedu
led for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  .not_valid_after(datetime.utcnow() + timedelta(days=365))
✅ Certificates generated successfully:
   - Private Key: /mnt/c/Users/User/Downloads/automation-main/automation-main/mockup-infra/certs/server.key
   - Certificate: /mnt/c/Users/User/Downloads/automation-main/automation-main/mockup-infra/certs/server.crt
   - Common Name: api.mockup.test
   - Subject Alt Names: localhost, api.mockup.test, 127.0.0.1, 172.18.0.1, 172.19.0.1

🌐 [Layer 3 - Network] Testing Isolation...
🔧 Executing: podman network inspect private_net 2>/dev/null || echo '{}'
[
    {
        "name": "private_net",
        "id": "036842daffb7cb908c8be6803908b8999bbe5053aabfc6c66f6a374abc938028",
        "driver": "bridge",
        "network_interface": "podman2",
        "created": "2026-03-05T20:12:04.108169658+07:00",
        "subnets": [
            {
                "subnet": "10.89.1.0/24",
                "gateway": "10.89.1.1"
            }
        ],
        "ipv6_enabled": false,
        "internal": true,
        "dns_enabled": true,
        "ipam_options": {
            "driver": "host-local"
        }
    }
]
✅ Private network is isolated (internal: true)
```

```
🚀 Deploying Mockup Infrastructure Stack
   Bare Metal → L3 → L4 → L5/6 → L7
==================================================

🌐 [Layer 3 - Network] Testing Isolation...
🔧 Executing: podman network inspect private_net 2>/dev/null || echo '{}'
[
    {
        "name": "private_net",
        "id": "036842daffb7cb908c8be6803908b8999bbe5053aabfc6c66f6a374abc938028",
        "driver": "bridge",
        "network_interface": "podman2",
        "created": "2026-03-05T20:12:04.108169658+07:00",
        "subnets": [
            {
                "subnet": "10.89.1.0/24",
                "gateway": "10.89.1.1"
            }
        ],
        "ipv6_enabled": false,
        "internal": true,
        "dns_enabled": true,
        "ipam_options": {
            "driver": "host-local"
        }
    }
]
✅ Private network is isolated (internal: true)
```

ใช้ Podman network ในการทดลอง และถูกสร้างแล้ว
ซึ่ง Subnet = 10.89.1.0/24 และIsolation เปิดอยู่

🟡 VERIFY CONTAINERS

```
None
podman ps
ใช้โปรแกรม Docker
-> docker ps
```

Should show:

Container	Port	Result
stateless-api	3000	✅
stateful-api	3001	✅

🧪 PART 3 — TESTING STATELESS vs STATEFUL (week03-stateless-stateful/phase1-mockup/README.md)

⚫ LOCAL TESTS (Private LAN)

From ClientA:

```shell
curl http://192.168.10.10:3000/info
```

→ Stateless

```
StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"Stateless server information","data":{"server":"Stateless Mock Server
                    v1.0","timestamp":"2026-03-05T12:50:54.109Z","requestCount":1,"clientId":"anonymous","randomValue":...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors
                    'self';img-src 'self' data:;object-src 'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors
                    'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https:
                    'unsafe-inline';upgrade-insecure-requests], [Cross-Origin-Opener-Policy, same-origin], [Cross-Origin-Resource-Policy, same-origin],
                    [Origin-Agent-Cluster, ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 422
```

จากการรันพบว่า

server ทำงานที่ http://localhost:3001
Response: HTTP 200
Stateless: requestCount = 1
clientId: "anonymous" นั่นคือ Client ไม่มี session

From ClientA:

```shell
curl -X POST http://192.168.10.10:3001/session
```

คำสั่งที่ใช้ในการสร้าง session

```shell
$rawResponse = curl -Method POST http://localhost:3002/session `
   -ContentType "application/json" `
   -Body '{"userId":"user001"}'
$rawResponse.Content
```

ผลที่ได้

```shell
{"status":"success","message":"Session created successfully","data":{"session":{"id":"909d1040-635e-4331-ba90-6fba0da94c6a","userId":"user001","createdAt":"2026-03-05T13:36:47.217Z","expiresAt":"2026-03-05T13:51:47.218Z"},"user":{"id":"user001","name":"Test User 1","email":"user1@example.com"},"note":"Session created. Include Session-ID header in subsequent requests to maintain state."},"timestamp":"2026-03-05T13:36:47.218Z"}
```

นั่นคือ Session Id : 909d1040-635e-4331-ba90-6fba0da94c6a

Store the returned `Session-ID`, then:

```shell
curl http://192.168.10.10:3001/data -H "Session-ID: <id>"
```

คำสั่งที่ใช้

```shell
curl -Headers @{"Session-ID"="909d1040-635e-4331-ba90-6fba0da94c6a"}
http://localhost:3002/profile
```

ผลลัพธ์

```
StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"User profile","data":{"profile":{"id":"user001","name":"Test User
                    1","email":"user1@example.com","preferences":{"theme":"light","language":"en"}},"note":"User profile re...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors
                    'self';img-src 'self' data:;object-src 'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors
                    'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https:
                    'unsafe-inline';upgrade-insecure-requests], [Cross-Origin-Opener-Policy, same-origin], [Cross-Origin-Resource-Policy, same-origin],
                    [Origin-Agent-Cluster, ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 313
```

mockup-infra deploy สำเร็จ
Gateway ส่งต่อ request ถูกต้อง
Network isolation ถูกตั้งค่าแล้ว

🔵 REMOTE TESTS (Via Simulated Internet)

เนื่องจากไม่มี infrastructure จึงใช้ Port Forwarding
โดยรันคำสั่ง บน cmd (as administrator)

```shell
netsh interface portproxy add v4tov4 listenaddress=10.10.12.2
listenport=3000 connectaddress=localhost connectport=3001

netsh interface portproxy add v4tov4 listenaddress=10.10.12.2
listenport=3001 connectaddress=localhost connectport=3002
```

```
C:\Windows\System32>netsh interface portproxy show all

Listen on ipv4:             Connect to ipv4:

Address         Port        Address         Port
--------------- ----------  --------------- ----------
10.10.12.2      3000        localhost       3001
10.10.12.2      3001        localhost       3002
```

จากภาพจะได้ว่า

10.10.12.2:3000 → forward ไป localhost:3001 (Stateless Server)
10.10.12.2:3001 → forward ไป localhost:3002 (Stateful Server)

ทดสอบ Stateless Remote

รันคำสั่ง บน cmd (as administrator)

```shell
netsh interface ip add address "Loopback" 10.10.12.2
255.255.255.252
```

และรันคำสั่งบน Powershell เพื่อทดสอบการเชื่อมต่อ

```shell
Test-NetConnection -ComputerName 10.10.12.2 -Port 3000
Test-NetConnection -ComputerName 10.10.12.2 -Port 3001
```

จะได้

```
ComputerName     : 10.10.12.2
RemoteAddress    : 10.10.12.2
RemotePort       : 3000
InterfaceAlias   : Loopback Pseudo-Interface 1
SourceAddress    : 10.10.12.2
TcpTestSucceeded : True
```

From ClientA:

```none
curl http://10.10.12.2:3000/info
```

That will target **ServerB stateless** — NAT + routing should deliver

จากการรันในผลลัพธ์ดังนี้

```
StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"Stateless server information","data":{"server":"Stateless Mock Server
                    v1.0","timestamp":"2026-03-05T14:23:52.210Z","requestCount":1,"clientId":"anonymous","randomValue":...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors
                    'self';img-src 'self' data:;object-src 'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors
                    'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https:
                    'unsafe-inline';upgrade-insecure-requests], [Cross-Origin-Opener-Policy, same-origin], [Cross-Origin-Resource-Policy, same-origin],
                    [Origin-Agent-Cluster, ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 422
```

สรุปได้ว่า Stateless Remote ทำงานได้ที่ `curl http://10.10.12.2:3000/info`

ทดสอบสร้าง Session ที่ Remote ด้วยคำสั่ง

```shell
$response = curl -Method POST http://10.10.12.2:3001/session `
  -ContentType "application/json" `
  -Body '{"userId":"user001"}' | ConvertFrom-Json


curl -Headers

@{"Session-ID"="95414afb-0f4c-414e-b16f-9a31ce40c8ab"} `

  http://10.10.12.2:3001/profile -Verbose
```

ได้ผลลัพธ์ดังนี้

```
StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"User profile","data":{"profile":{"id":"user001","name":"Test User
                    1","email":"user1@example.com","preferences":{"theme":"light","language":"en"}},"note":"User profile re...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors
                    'self';img-src 'self' data:;object-src 'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors
                    'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https:
                    'unsafe-inline';upgrade-insecure-requests], [Cross-Origin-Opener-Policy, same-origin], [Cross-Origin-Resource-Policy, same-origin],
                    [Origin-Agent-Cluster, ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 313
```

```shell
curl -Headers
@{"Session-ID"="95414afb-0f4c-414e-b16f-9a31ce40c8ab"} `
  http://10.10.12.2:3001/info
```

```
StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"Stateful server information","data":{"server":"Stateful Mock Server
                    v1.0","timestamp":"2026-03-05T14:40:18.737Z","sessionInfo":{"sessionId":"95414afb-0f4c-414e-b16f-9a31...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors
                    'self';img-src 'self' data:;object-src 'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors
                    'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https:
                    'unsafe-inline';upgrade-insecure-requests], [Cross-Origin-Opener-Policy, same-origin], [Cross-Origin-Resource-Policy, same-origin],
                    [Origin-Agent-Cluster, ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 500
```

Session Test:

```none
curl -X POST http://10.10.12.2:3001/session
```

→ ServerB returns a session ID

Then:

```none
curl http://10.10.12.2:3001/data -H "Session-ID: <id>"
```

Check session persists across NAT

ทดสอบ Shopping Cart (Stateful Feature) โดยใช้คำสั่ง

```shell
Write-Host "`n=== ทดสอบ Shopping Cart ผ่าน NAT ===" -ForegroundColor
Cyan

# เพิ่มสินค้า
curl -Method POST http://10.10.12.2:3001/cart/add `
```

```
  -Headers @{"Session-ID"=$remoteSessionId;
"Content-Type"="application/json"} `
  -Body '{"productId":"prod001","quantity":2}'

# ดูสินค้าใน cart
curl -Headers @{"Session-ID"=$remoteSessionId} `
  http://10.10.12.2:3001/cart

# 1. เพิ่มสินค้าชิ้นที่ 2
Write-Host "เพิ่มสินค้าชิ้นที่ 2..." -ForegroundColor Cyan
curl -Method POST http://10.10.12.2:3001/cart/add `
  -Headers @{"Session-ID"=$remoteSessionId;
"Content-Type"="application/json"} `
  -Body '{"productId":"prod002","quantity":1}'

# 2. ดู cart (ควรมี 2 รายการ)
Write-Host "`nดู cart อีกครั้ง..." -ForegroundColor Cyan
curl -Headers @{"Session-ID"=$remoteSessionId}
http://10.10.12.2:3001/cart

# 3. ดูข้อมูล session
Write-Host "`nดูข้อมูล session..." -ForegroundColor Cyan
curl -Headers @{"Session-ID"=$remoteSessionId}
http://10.10.12.2:3001/session

# 4. ทดสอบ stateful demonstration
Write-Host "`nทดสอบ stateful demonstration..." -ForegroundColor Cyan
curl -Headers @{"Session-ID"=$remoteSessionId}
http://10.10.12.2:3001/demonstrate/stateful
```

ผลลัพธ์

Shell

```
เพิ่มสินค้าใน cart...


StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"Item added to cart","data"
                    :{"cart":{"sessionId":"59adf81d-ee7b-41db-b095-8d2c07192a
                    98","items":[{"productId":"prod001","name":"Sample
                    Product A","price":29.99,"quantity"...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri
                    'self';font-src 'self' https: data:;form-action                                'self';frame-ancestors 'self';img-src
'self'                                data:;object-src 'none';script-src 's...                 Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri
                    'self';font-src 'self' https: data:;form-action
                    'self';frame-ancestors 'self';img-src 'self'
                    data:;object-src 'none';script-src
                    'self';script-src-attr 'none';style-src 'self' https:
                    'unsafe-inline';upgrade-insecure-requests],
                    [Cross-Origin-Opener-Policy, same-origin],
                    [Cross-Origin-Resource-Policy, same-origin],
                    [Origin-Agent-Cluster, ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 486


ดูสินค้าใน cart...
StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"Cart retrieved","data":{"c
                    art":{"sessionId":"59adf81d-ee7b-41db-b095-8d2c07192a98",
                    "items":[{"productId":"prod001","name":"Sample Product
                    A","price":29.99,"quantity":2,"...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri
                    'self';font-src 'self' https: data:;form-action
                    'self';frame-ancestors 'self';img-src 'self'
                    data:;object-src 'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri
                    'self';font-src 'self' https: data:;form-action
                    'self';frame-ancestors 'self';img-src 'self'
                    data:;object-src 'none';script-src
                    'self';script-src-attr 'none';style-src 'self' https:
                    'unsafe-inline';upgrade-insecure-requests],
                    [Cross-Origin-Opener-Policy, same-origin],
                    [Cross-Origin-Resource-Policy, same-origin],
                    [Origin-Agent-Cluster, ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 496



เพิ่มสินค้าชิ้นที่ 2...

StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"Item added to

cart","data":{"cart":{"sessionId":"59adf81d-ee7b-41db-b095-8d2c07192a98","items":[{"productId":"prod001","name":"Sample Product
A","price":29.99,"quantity"...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action
'self';frame-ancestors 'self';img-src 'self' data:;object-src
                    'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action
'self';frame-ancestors 'self';img-src 'self'
                    data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https:
'unsafe-inline';upgrade-insecure-requests], [Cross-Origin-Opener-Policy,
                    same-origin], [Cross-Origin-Resource-Policy, same-origin], [Origin-Agent-Cluster, ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 600
```

```
ดู cart อีกครั้ง...
StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"Cart
retrieved","data":{"cart":{"sessionId":"59adf81d-ee7b-41db-b095-8d2c07192a98","items":[{"productId":"prod001","name":"Sample
                    Product A","price":29.99,"quantity":2,"...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action
'self';frame-ancestors 'self';img-src 'self' data:;object-src
                    'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action
'self';frame-ancestors 'self';img-src 'self'
                    data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https:
'unsafe-inline';upgrade-insecure-requests], [Cross-Origin-Opener-Policy,
                    same-origin], [Cross-Origin-Resource-Policy, same-origin], [Origin-Agent-Cluster, ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 610


ดูข้อมูล session...
StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"Session

information","data":{"session":{"id":"59adf81d-ee7b-41db-b095-8d2c07192a98","userId":"user001","createdAt":"2026-03-05T14:46:37.160Z","las
tAccessed":"2026-03-05T...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action
'self';frame-ancestors 'self';img-src 'self' data:;object-src
                    'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action
'self';frame-ancestors 'self';img-src 'self'
                    data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https:
'unsafe-inline';upgrade-insecure-requests], [Cross-Origin-Opener-Policy,
                    same-origin], [Cross-Origin-Resource-Policy, same-origin], [Origin-Agent-Cluster, ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 385


ทดสอบ stateful demonstration...
StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"Stateful behavior demonstration","data":{"demonstration":"Stateful Server
                    Behavior","sessionInfo":{"sessionId":"59adf81d-ee7b-41db-b095-8d2c07192a98","userId":"user001",...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action
'self';frame-ancestors 'self';img-src 'self' data:;object-src
                    'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action
'self';frame-ancestors 'self';img-src 'self'
                    data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https:
'unsafe-inline';upgrade-insecure-requests], [Cross-Origin-Opener-Policy,
                    same-origin], [Cross-Origin-Resource-Policy, same-origin], [Origin-Agent-Cluster, ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 790
```

สรุปผลการทดสอบ

Session ID: 59adf81d-ee7b-41db-b095-8d2c07192a98

1. เพิ่มสินค้าชิ้นที่ 2: สำเร็จ
   - สินค้า: prod002 จำนวน 1 ชิ้น

2. ดู Cart อีกครั้ง: สำเร็จ
   - สินค้าใน cart: 2 รายการ
   - รายการที่ 1: prod001 x2 = $59.98
   - รายการที่ 2: prod002 x1 = $19.99
   - Total: $79.97

3. ดูข้อมูล Session: สำเร็จ
   - Session ID: 59adf81d-ee7b-41db-b095-8d2c07192a98
   - User ID: user001
   - Created: 2026-03-05T14:46:37.160Z
   - Last Accessed: อัปเดตล่าสุด

4. Stateful Demonstration: สำเร็จ
   - Server จำ session ได้
   - Visit count นับเพิ่ม

⚠️ TROUBLESHOOTING

❗ Cannot Reach Remote

- กรณี Gateway misconfiguration
  รันโค้ดใน cmd(aministrator)

```shell
#ตรวจสอบ route ปัจจุบัน

route print

# ลบ route ส่งผลให้ gateway ผิด

route delete 10.10.12.0

# ทดสอบ ping

ping 10.10.12.2
```

```
C:\Windows\System32>route delete 10.10.12.2
 OK!

C:\Windows\System32>ping 10.10.12.2

Pinging 10.10.12.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.12.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

ผลลัพธ์คือ ไม่สามารถ ping ไปที่ 10.10.12.2 ได้

- NAT ACL incorrect

```shell
netsh interface portproxy add v4tov4 listenaddress=10.10.12.2
listenport=3000 connectaddress=localhost connectport=3002

netsh interface portproxy add v4tov4 listenaddress=10.10.12.2
listenport=3001 connectaddress=localhost connectport=3001

netsh interface portproxy show all
```

```
Listen on ipv4:             Connect to ipv4:

Address         Port        Address         Port
--------------- ----------  --------------- ----------
10.10.12.2      3000        localhost       3002
10.10.12.2      3001        localhost       3001
```

จากภาพ คือทำการสลับ port ระหว่าง Stateless (port 3000) และ Stateful (port 3001) เรียบร้อย

ทดสอบ Stateless Remote

```
C:\Windows\System32>curl http://10.10.12.2:3000/info
curl: (28) Failed to connect to 10.10.12.2 port 3000 after 21337 ms: Could not connect to server

C:\Windows\System32>curl -X POST http://10.10.12.2:3001/session -H "Content-Type: application/json" -d "{\"userId\":\"user001\"}"
curl: (28) Failed to connect to 10.10.12.2 port 3001 after 21438 ms: Could not connect to server

C:\Windows\System32>telnet 10.10.12.2 3000
Connecting To 10.10.12.2...Could not open connection to the host, on port 3000: Connect failed
```

ผลลัพธ์จากการทดสอบ Stateless Remote สร้าง session และ telnet สรุปได้ว่า fail to connect

- Wrong default gateway on clients

Check:

```
None
show ip route
ping 10.10.12.2 from R1
```

❗ Sessions Lose State

- Missing header
- Wrong container port
- Firewall blocking port 3001